

Machine Learning – Lecture 1

Introduction to Supervised Machine Learning (Theory up to Train-Test Split)



Course Resources

1. Google Colab Notebook (Hands-on Practice)

The practical implementation for this course will be carried out using Google Colab. Please bookmark the notebook and keep it ready for the next class.

Google Colab Notebook:

<https://colab.research.google.com/drive/1FNQR7wFEfFH7a9cHIntupTWIRezSM4-J#scrollTo=AXqVh8jl4JD0>

Google Colab is a free cloud-based Python notebook provided by Google that allows us to write and execute Python code directly in the browser without installing any software. It is one of the most widely used environments for Machine Learning and Data Science.

2. Class Notes

Detailed lecture notes are available here:

<https://1drv.ms/o/c/350d1dd72dbeac65/IgDi5Nr3-zYyTK3X7UAP8yjPAZevZVF53w6RmvpPb7j6YG4?e=BKzkkr>

Please revise these notes before the next lecture.

3. Kaggle Learning Resource

Students who wish to explore a practical implementation after the theory sessions can refer to the following Kaggle notebook.

<https://www.kaggle.com/code/ashydv/sales-prediction-simple-linear-regression>

Note: We will understand every line of this notebook during the practical sessions. There is no need to memorize the code now.

4. Google Teachable Machine (No-Code Machine Learning)

Google Teachable Machine

<https://teachablemachine.withgoogle.com/>

Google Teachable Machine is a free web-based tool developed by Google that allows anyone to create Machine Learning models **without writing a single line of code**.

It is one of the best tools for beginners to understand how Machine Learning works.

What can you build?

- Image Classification Models
- Audio Classification Models
- Pose Detection Models

Why are we using it?

Before writing Python code, it is important to understand **how a machine actually learns**.

With Google Teachable Machine, you can:

- Collect training examples
- Train a Machine Learning model with one click
- Test the trained model instantly
- Understand the concepts of **training**, **prediction**, and **classification** visually

This helps students understand the Machine Learning workflow before moving on to Scikit-learn and Python programming.

Course Outcomes

After today's lecture, students should be able to

- ✓ Understand Artificial Intelligence
 - ✓ Understand Machine Learning
 - ✓ Differentiate between Traditional Programming and Machine Learning
 - ✓ Understand the different types of Machine Learning
 - ✓ Explain Supervised Learning
 - ✓ Differentiate Regression and Classification
 - ✓ Identify Features and Target Variables
 - ✓ Understand the need for Train-Test Split
-

1. What is Artificial Intelligence?

Artificial Intelligence (AI) is the branch of computer science that enables computers to perform tasks that normally require human intelligence.

Examples include

- ChatGPT
- Siri
- Google Assistant
- Face Unlock
- Netflix Recommendations
- Self Driving Cars

AI is the broadest field.

Machine Learning is a subset of Artificial Intelligence.

2. What is Machine Learning?

Machine Learning is a technique where computers learn patterns from data instead of being explicitly programmed.

Instead of writing rules,

we provide

- Data
- Correct Answers

and the machine learns the relationship automatically.

Traditional Programming

Input

+

Rules

↓

Output

Example

If Marks > 40

Pass

Else

Fail

Here the programmer writes all the rules.

Machine Learning

Input Data

+

Correct Answers

↓

Machine Learns

↓

Prediction

Example

Hours Studied	Marks
2	30
4	50
6	70
8	90

The computer learns

More Study Hours

↓

Higher Marks

without explicitly writing any rule.

3. Types of Machine Learning

Machine Learning is mainly divided into three categories.

Machine Learning

|

|— Supervised Learning

— Unsupervised Learning

— Reinforcement Learning

Supervised Learning

The dataset already contains the correct answers.

Example

Area	House Price
1000	50 Lakhs
1500	70 Lakhs

The machine learns how Area affects House Price.

4. What is Supervised Learning?

The word "Supervised" means there is a teacher.

The teacher already knows the correct answers.

Example

TV Advertisement	Sales
100	10
200	18
300	28

The machine learns

Advertisement

↓

Sales

This is called Supervised Learning because the output is already known.

5. Types of Supervised Learning

There are two major categories.

Supervised Learning

|

|— Regression

|— Classification

Regression

Regression predicts numerical values.

Examples

- House Price
- Salary
- Temperature
- Sales
- Rainfall

Example Output

Predicted Sales = 23.75

The output is always a number.

Classification

Classification predicts categories.

Examples

Spam / Not Spam

Healthy / Sick

High Sales / Low Sales

Male / Female

Pass / Fail

Example Output

High Sales

The output is a category.

6. Understanding the Dataset

We will use the Advertising Dataset.

It contains four columns.

Column	Description
TV	Money spent on TV Advertisements
Radio	Money spent on Radio Advertisements
Newspaper	Money spent on Newspaper Advertisements
Sales	Total Product Sales

7. Features and Target Variable

Every Machine Learning problem contains

Input Variables

↓

Features (X)

Output Variable

↓

Target (Y)

For our dataset

Features

TV

Radio

Newspaper

Target

Sales

8. Why Do We Split the Dataset?

Imagine a teacher preparing students for an examination.

Suppose there are 100 questions.

If students practice all 100 questions and the examination also contains the same 100 questions,

getting 100 marks does not prove that they actually learned.

Instead,

Teacher teaches

80 Questions

↓

Exam contains

20 New Questions

Now the marks truly indicate whether the students learned the concepts.

Machine Learning follows exactly the same principle.

9. Train-Test Split

The dataset is divided into

Entire Dataset

↓

Training Data (80%)

↓

Machine Learns

↓

Testing Data (20%)

↓

Model Performance

Training Dataset

Used to teach the model.

The model finds relationships between inputs and outputs.

Testing Dataset

Used only after training.

The model predicts outputs for unseen data.

This tells us how well the model performs.

10. Why Not Use 100% Training Data?

If the model learns every example,

it may simply memorize the dataset.

It may fail when new data is given.

This problem is known as **Overfitting**, which we will study later in the course.

11. Summary

After today's lecture you should understand

- Artificial Intelligence
 - Machine Learning
 - Traditional Programming
 - Machine Learning Workflow
 - Types of Machine Learning
 - Supervised Learning
 - Regression
 - Classification
 - Features
 - Target Variable
 - Dataset
 - Train-Test Split
-



Recommended Learning Videos

1. freeCodeCamp.org

Machine Learning for Beginners

https://www.youtube.com/results?search_query=freecodecamp+machine+learning+for+beginners

2. CodeBasics

Machine Learning Playlist

https://www.youtube.com/results?search_query=codebasics+machine+learning

3. Krish Naik

Machine Learning Playlist

https://www.youtube.com/results?search_query=krish+naik+machine+learning

4. StatQuest with Josh Starmer

Machine Learning Basics

https://www.youtube.com/results?search_query=statquest+machine+learning

5. Simplilearn

Introduction to Machine Learning

https://www.youtube.com/results?search_query=simplilearn+machine+learning



Assignment – 1

Course: Machine Learning

Topics Covered:

- Artificial Intelligence
- Machine Learning

- Types of Machine Learning
 - Supervised Learning
 - Regression
 - Classification
 - Features
 - Target Variable
 - Advertising Dataset
 - Train-Test Split
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Part A – Multiple Choice Questions (1 Mark Each)

1. Machine Learning is a subset of

- a) Database
 - b) Artificial Intelligence
 - c) Networking
 - d) Operating System
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2. Which type of Machine Learning uses labelled data?

- a) Reinforcement Learning
 - b) Supervised Learning
 - c) Unsupervised Learning
 - d) Deep Learning
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3. Which of the following is a Regression problem?

- a) Email Spam Detection
- b) House Price Prediction
- c) Face Recognition
- d) Disease Detection

4. Which of the following is a Classification problem?

- a) Temperature Prediction
 - b) Sales Prediction
 - c) House Price Prediction
 - d) Email Spam Detection
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5. Which column is the Target Variable in the Advertising Dataset?

- a) TV
 - b) Radio
 - c) Newspaper
 - d) Sales
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Part B – Fill in the Blanks

1. Machine Learning is a subset of _____.
 2. Regression predicts _____ values.
 3. Classification predicts _____.
 4. Input variables are called _____.
 5. Output variable is called the _____.
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Part C – Short Answer Questions

1. Define Artificial Intelligence.
2. What is Machine Learning?
3. Explain the difference between Traditional Programming and Machine Learning.
4. What is Supervised Learning? Give one real-world example.
5. Differentiate between Regression and Classification with examples.
6. What are Features and Target Variables in the Advertising Dataset?
7. Why do we perform Train-Test Split?

Part D – Dataset Understanding

The Advertising Dataset contains the following columns:

- TV
- Radio
- Newspaper
- Sales

Answer the following:

1. Identify the input features.
 2. Identify the target variable.
 3. Why should we not train the model using the complete dataset?
 4. What is the purpose of the Testing dataset?
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Bonus Question (2 Marks)

Draw the following Machine Learning workflow neatly and explain each stage.

Data

↓

Training

↓

Model

↓

Prediction

Preparation for the Next Class

Before attending the next lecture, students are expected to:

- Read the lecture notes.
- Open the Google Colab notebook and familiarize themselves with the interface.
- Watch at least one of the recommended videos.
- Explore the Kaggle notebook (only observe the workflow; detailed coding will be covered in class).
- Complete and submit Assignment 1.

In the next lecture, we will begin **hands-on implementation** using the Advertising dataset in Google Colab, covering dataset loading, data exploration, feature selection, target selection, train-test split, and then building our first **Linear Regression** model step by step with a detailed explanation of every line of code.